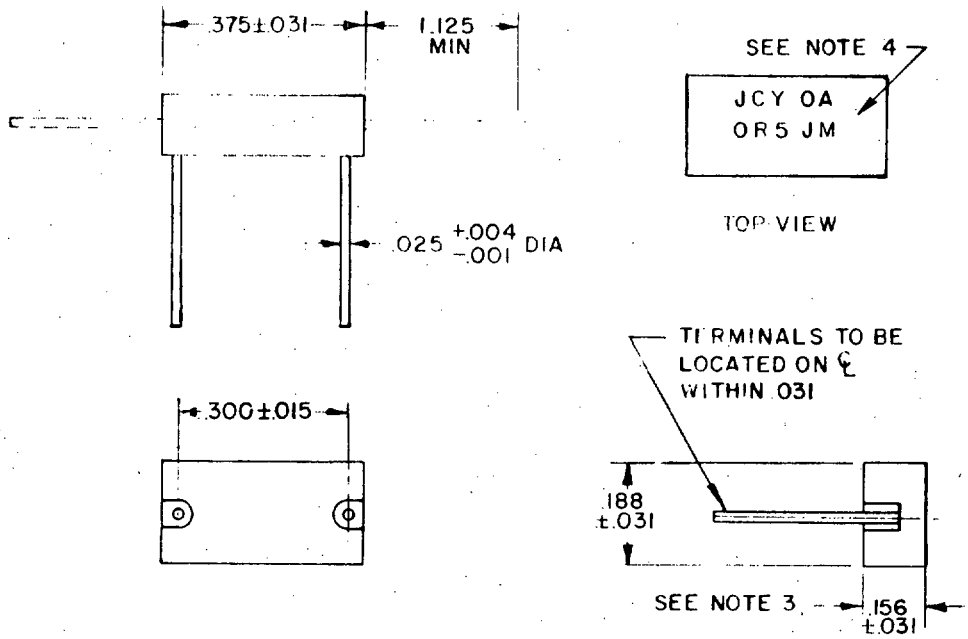
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, GLASS DIELECTRIC, (AXIAL-RADIAL WIRE-LEAD TERMINALS), ESTABLISHED RELIABILITY, STYLE CYR13

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-23269.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



INCHES	MM	INCHES	MM
.001	.03	.188	4.78
.004	.10	.219	5.56
.015	.38	.300	7.62
.025	.64	.375	9.53
.031	.79	1.125	28.58
.156	3.96		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. For any value above 130 pF this dimension shall be .219 ±.031.
4. See marking requirement.

FSC 5910

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:14Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure.
Case material - Glass (vitreous enamel).
Terminals - Type C of MIL-STD-1276.
Capacitance (Cap.) (nom) - See table I.
Cap. tolerance - See table I.
Rated voltage - See table I.
Failure rate level - L, M, P, R, or S.
Rated temperature - -55° to +125°C.
Temperature coefficient - 105 ±25 parts/million/°C.
Capacitance drift - 0.1% or 0.1 pF, whichever is greater.
Seal - Hermetic.

Seal: Test II.

Insulation resistance: 100,000 megohms or 200 ohm-farads min (qualification inspection only at +125°C). 500,000 megohms min (at +25°C).

Dissipation factor (DF): 0.7% max (0.5 thru 4.7 pF); 0.3% max (5.1 thru 27 pF); and 0.1% max (30 pF and up).

Thermal shock and moisture resistance:
DF - Not to exceed initial requirement.

Q: 2,000 min.

Life:
ΔCap (240 hr and 2,000 hr) - 1.0% or 0.5 pF, max.
DF (240 hr and 2,000 hr) - Not to exceed initial requirement.

Marking:

"J" Brand	Capacitor indicator	Dielectric indicator	Last digit of year code	Sequential 4-week lot code
J	C	Y	0	A
O	R	5	J	M
	Capacitance code		Capacitance tolerance code	Failure rate level code (%/1,000 hrs)

Capacitance tolerance code:
C = ±0.25 pF
D = ±0.50 pF
F = ±1%
G = ±2%
J = ±5%

Failure rate level code (%/1,000 hrs):
L = 2.0
M = 1.0
P = 0.1
R = 0.01
S = 0.001

Substitutability data: See table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:14Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical and physical characteristics and dash numbers.

Cap.	Rated voltage	Capacitance tolerance	Dash number M23269/05- FR level in %/1,000 hours				
			L (2.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
(pF)	(volts, dc)						
0.5	500	±0.25 pF	2001	3001	4001	5001	6001
1.0		±0.25 pF	2002	3002	4002	5002	6002
1.5		±0.25 pF	2003	3003	4003	5003	6003
2.2		±0.25 pF	2004	3004	4004	5004	6004
2.2		±0.50 pF	2005	3005	4005	5005	6005
2.7		±0.25 pF	2006	3006	4006	5006	6006
3.0		±0.25 pF	2007	3007	4007	5007	6007
3.0		±0.50 pF	2008	3008	4008	5008	6008
3.3		±0.25 pF	2009	3009	4009	5009	6009
3.6		±0.25 pF	2010	3010	4010	5010	6010
3.6		±0.50 pF	2011	3011	4011	5011	6011
3.9		±0.25 pF	2012	3012	4012	5012	6012
4.3		±0.25 pF	2013	3013	4013	5013	6013
4.3		±0.50 pF	2014	3014	4014	5014	6014
4.7		±0.25 pF	2015	3015	4015	5015	6015
5.1		±0.25 pF	2016	3016	4016	5016	6016
5.6		±0.25 pF	2017	3017	4017	5017	6017
5.6		±5%	2018	3018	4018	5018	6018
6.2		±0.25 pF	2019	3019	4019	5019	6019
6.2		±5%	2020	3020	4020	5020	6020
6.8		±0.25 pF	2021	3021	4021	5021	6021
6.8		±5%	2022	3022	4022	5022	6022
7.5		±0.25 pF	2023	3023	4023	5023	6023
7.5		±5%	2024	3024	4024	5024	6024
8.2		±0.25 pF	2025	3025	4025	5025	6025
8.2		±5%	2026	3026	4026	5026	6026
9.1		±0.25 pF	2027	3027	4027	5027	6027
9.1		±5%	2028	3028	4028	5028	6028
10		±0.25 pF	2029	3029	4029	5029	6029
10		±5%	2030	3030	4030	5030	6030
11		±0.25 pF	2031	3031	4031	5031	6031
11		±5%	2032	3032	4032	5032	6032
12		±0.25 pF	2033	3033	4033	5033	6033
12		±5%	2034	3034	4034	5034	6034
13		±2%	2035	3035	4035	5035	6035
13		±5%	2036	3036	4036	5036	6036
15		±2%	2037	3037	4037	5037	6037
15		±5%	2038	3038	4038	5038	6038
16		±2%	2039	3039	4039	5039	6039
16		±5%	2040	3040	4040	5040	6040
18		±2%	2041	3041	4041	5041	6041
18		±5%	2042	3042	4042	5042	6042
20		±2%	2043	3043	4043	5043	6043
20		±5%	2044	3044	4044	5044	6044
22		±2%	2045	3045	4045	5045	6045
22		±5%	2046	3046	4046	5046	6046
24		±2%	2047	3047	4047	5047	6047
24		±5%	2048	3048	4048	5048	6048
27		±1%	2049	3049	4049	5049	6049
27		±2%	2050	3050	4050	5050	6050
27	500	±5%	2051	3051	4051	5051	6051

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:14Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical and physical characteristics and dash numbers - Continued

Cap.	Rated voltage	Capacitance tolerance	Dash number M23269/05- FR level in %/1,000 hours				
			L (2.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
(pF)	(volts, dc)						
30	500	±1%	2052	3052	4052	5052	6052
30	↑	±2%	2053	3053	4053	5053	6053
30		±5%	2054	3054	4054	5054	6054
33		±1%	2055	3055	4055	5055	6055
33		±2%	2056	3056	4056	5056	6056
33		±5%	2057	3057	4057	5057	6057
36		±1%	2058	3058	4058	5058	6058
36		±2%	2059	3059	4059	5059	6059
36		±5%	2060	3060	4060	5060	6060
39		±1%	2061	3061	4061	5061	6061
39		±2%	2062	3062	4062	5062	6062
39		±5%	2063	3063	4063	5063	6063
43		±1%	2064	3064	4064	5064	6064
43		±2%	2065	3065	4065	5065	6065
43		±5%	2066	3066	4066	5066	6066
47		±1%	2067	3067	4067	5067	6067
47		±2%	2068	3068	4068	5068	6068
47		±5%	2069	3069	4069	5069	6069
51		±1%	2070	3070	4070	5070	6070
51		±2%	2071	3071	4071	5071	6071
51		±5%	2072	3072	4072	5072	6072
56		±1%	2073	3073	4073	5073	6073
56		±2%	2074	3074	4074	5074	6074
56		±5%	2075	3075	4075	5075	6075
62		±1%	2076	3076	4076	5076	6076
62		±2%	2077	3077	4077	5077	6077
62		±5%	2078	3078	4078	5078	6078
68		±1%	2079	3079	4079	5079	6079
68		±2%	2080	3080	4080	5080	6080
68		±5%	2081	3081	4081	5081	6081
75		±1%	2082	3082	4082	5082	6082
75		±2%	2083	3083	4083	5083	6083
75		±5%	2084	3084	4084	5084	6084
82		±1%	2085	3085	4085	5085	6085
82		±2%	2086	3086	4086	5086	6086
82		±5%	2087	3087	4087	5087	6087
91		±1%	2088	3088	4088	5088	6088
91		±2%	2089	3089	4089	5089	6089
91		±5%	2090	3090	4090	5090	6090
100		±1%	2091	3091	4091	5091	6091
100		±2%	2092	3092	4092	5092	6092
100		±5%	2093	3093	4093	5093	6093
110		±1%	2094	3094	4094	5094	6094
110		±2%	2095	3095	4095	5095	6095
110		±5%	2096	3096	4096	5096	6096
120		±1%	2097	3097	4097	5097	6097
120	↓	±2%	2098	3098	4098	5098	6098
120	500	±5%	2099	3099	4099	5099	6099

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:14Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical and physical characteristics and dash numbers. - Continued

Cap.	Rated voltage	Capacitance tolerance	Dash number M23269/05- FR level in %/1,000 hours				
			L (2.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
(pF)	(volts, dc)						
130	500	±1%	2100	3100	4100	5100	6100
130		±2%	2101	3101	4101	5101	6101
130		±5%	2102	3102	4102	5102	6102
150		±1%	2103	3103	4103	5103	6103
150		±2%	2104	3104	4104	5104	6104
150		±5%	2105	3105	4105	5105	6105
160		±1%	2106	3106	4106	5106	6106
160		±2%	2107	3107	4107	5107	6107
160		±5%	2108	3108	4108	5108	6108
180		±1%	2109	3109	4109	5109	6109
180		±2%	2110	3110	4110	5110	6110
180		±5%	2111	3111	4111	5111	6111
200		±1%	2112	3112	4112	5112	6112
200		±2%	2113	3113	4113	5113	6113
200	500	±5%	2114	3114	4114	5114	6114
220	300	±1%	2115	3115	4115	5115	6115
220		±2%	2116	3116	4116	5116	6116
220		±5%	2117	3117	4117	5117	6117
240		±1%	2118	3118	4118	5118	6118
240		±2%	2119	3119	4119	5119	6119
240		±5%	2120	3120	4120	5120	6120
270		±1%	2121	3121	4121	5121	6121
270		±2%	2122	3122	4122	5122	6122
270		±5%	2123	3123	4123	5123	6123
300		±1%	2124	3124	4124	5124	6124
300		±2%	2125	3125	4125	5125	6125
300	300	±5%	2126	3126	4126	5126	6126

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:14Z
Check the source to verify that this is the current version before use.

TABLE II. Substitutability data.

Superseded type designation in MIL-C-23269/5B	Use dash number M23269/05- in MIL-C-23269/5C FR level in %/1, 000 hrs		Superseded type designation in MIL-C-23269/5B	Use dash number M23269/05- in MIL-C-23269/5C FR level in %/1, 000 hrs	
	L (2.0)	M (1.0)		L (2.0)	M (1.0)
CYR13D0R5C	2001	3001	CYR13D300F	2052	3052
CYR13D1R0C	2002	3002	CYR13D300G	2053	3053
CYR13D1R5C	2003	3003	CYR13D300J	2054	3054
CYR13D2R2C	2004	3004	CYR13D330F	2055	3055
CYR13D2R2D	2005	3005	CYR13D330G	2056	3056
CYR13D2R7C	2006	3006	CYR13D330J	2057	3057
CYR13D3R0C	2007	3007	CYR13D360F	2058	3058
CYR13D3R0D	2008	3008	CYR13D360G	2059	3059
CYR13D3R3C	2009	3009	CYR13D360J	2060	3060
CYR13D3R6C	2010	3010	CYR13D390F	2061	3061
CYR13D3R6D	2011	3011	CYR13D390G	2062	3062
CYR13D3R9C	2012	3012	CYR13D390J	2063	3063
CYR13D4R3C	2013	3013	CYR13D430F	2064	3064
CYR13D4R3D	2014	3014	CYR13D430G	2065	3065
CYR13D4R7C	2015	3015	CYR13D430J	2066	3066
CYR13D5R1C	2016	3016	CYR13D470F	2067	3067
CYR13D5R6C	2017	3017	CYR13D470G	2068	3068
CYR13D5R6J	2018	3018	CYR13D470J	2069	3069
CYR13D6R2C	2019	3019	CYR13D510F	2070	3070
CYR13D6R2J	2020	3020	CYR13D510G	2071	3071
CYR13D6R8C	2021	3021	CYR13D510J	2072	3072
CYR13D6R8J	2022	3022	CYR13D560F	2073	3073
CYR13D7R5C	2023	3023	CYR13D560G	2074	3074
CYR13D7R5J	2024	3024	CYR13D560J	2075	3075
CYR13D8R2C	2025	3025	CYR13D620F	2076	3076
CYR13D8R2J	2026	3026	CYR13D620G	2077	3077
CYR13D9R1C	2027	3027	CYR13D620J	2078	3078
CYR13D9R1J	2028	3028	CYR13D680F	2079	3079
CYR13D100C	2029	3029	CYR13D680G	2080	3080
CYR13D100J	2030	3030	CYR13D680J	2081	3081
CYR13D110C	2031	3031	CYR13D750F	2082	3082
CYR13D110J	2032	3032	CYR13D750G	2083	3083
CYR13D120C	2033	3033	CYR13D750J	2084	3084
CYR13D120J	2034	3034	CYR13D820F	2085	3085
CYR13D130G	2035	3035	CYR13D820G	2086	3086
CYR13D130J	2036	3036	CYR13D820J	2087	3087
CYR13D150G	2037	3037	CYR13D910F	2088	3088
CYR13D150J	2038	3038	CYR13D910G	2089	3089
CYR13D160G	2039	3039	CYR13D910J	2090	3090
CYR13D160J	2040	3040	CYR13D101F	2091	3091
CYR13D180G	2041	3041	CYR13D101G	2092	3092
CYR13D180J	2042	3042	CYR13D101J	2093	3093
CYR13D200G	2043	3043	CYR13D111F	2094	3094
CYR13D200J	2044	3044	CYR13D111G	2095	3095
CYR13D220G	2045	3045	CYR13D111J	2096	3096
CYR13D220J	2046	3046	CYR13D121F	2097	3097
CYR13D240G	2047	3047	CYR13D121G	2098	3098
CYR13D240J	2048	3048	CYR13D121J	2099	3099
CYR13D270F	2049	3049	CYR13D131F	2100	3100
CYR13D270G	2050	3050	CYR13D131G	2101	3101
CYR13D270J	2051	3051	CYR13D131J	2102	3102

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:14Z
Check the source to verify that this is the current version before use.

TABLE II. Substitutability data. - Continued

Superseded type designation in MIL-C-23269/5B	Use dash number M23269/05- in MIL-C-23269/5C FR level in %/1, 000 hrs		Superseded type designation in MIL-C-23269/5B	Use dash number M23269/05- in MIL-C-23269/5C FR level in %/1, 000 hrs	
	L (2.0)	M (1.0)		L (2.0)	M (1.0)
CYR13D151F	2103	3103	CYR13D221F	2115	3115
CYR13D151G	2104	3104	CYR13D221G	2116	3116
CYR13D151J	2105	3105	CYR13D221J	2117	3117
CYR13D161F	2106	3106	CYR13D241F	2118	3118
CYR13D161G	2107	3107	CYR13D241G	2119	3119
CYR13D161J	2108	3108	CYR13D241J	2120	3120
CYR13D181F	2109	3109	CYR13D271F	2121	3121
CYR13D181G	2110	3110	CYR13D271G	2122	3122
CYR13D181J	2111	3111	CYR13D271J	2123	3123
CYR13D201F	2112	3112	CYR13D301F	2124	3124
CYR13D201G	2113	3113	CYR13D301G	2125	3125
CYR13D201J	2114	3114	CYR13D301J	2126	3126

Custodians:
Army - EL
Navy - EC
Air Force - 17

Review activities:
Army - EL
Navy -
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - None
Navy - MC, CG, AS, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5910-1008-5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:14Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:14Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:14Z
Check the source to verify that this is the current version before use.